NORMAN BUI

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Education

Carleton University | *Bachelor of Computer Science (Honours),* ***3.9/4.0 GPA*** Graduated Spring 2026

- Coursework: Statistical Modeling, Data Structures & Algorithms, Database Management Systems, Linear Algebra
- Awards: Dean's Honour List every year; President's, Henry Marshall Tory, and Harry S. Southam Scholarships

Experience

Canadian Tire Corporation | *Software Developer* July 2026 - Present

- Built an internal tool that turns a service's source code into written documentation
- Grounded every page in facts from the code; a claim-by-claim audit found **1.9%** of **1,694** claims wrong or misleading
- Cut the cost of each run **38x** by testing candidate AI models on real projects, with no loss in accuracy
- Built an AI grader calibrated against human grades, and used it to A/B test **4** designs on **4** real projects

Electronic Arts (EA) | *Software Developer Intern* May 2025 - Aug 2025

- Raised automated performance test pass rates from **35% to 100%** by fixing broken logins and setup steps
- Evaluated an AI system that plays the game to find bugs, measuring how reliably it worked
- Catalogued its failure patterns and wrote up findings the team used to improve it

Autodesk | *Software Developer Intern* Jan 2025 - Apr 2025

- Built systems that find and fix missing manufacturing data in Autodesk Fusion, with a repair step **8x** faster
- Designed predictable, rule-based checks for repairing manufacturing data
- Shipped services that check and repair customer data automatically, with **100%** test coverage

Public Health Agency of Canada | *Data Scientist Intern* Sep 2024 - Dec 2024

- Built an AI-assisted research review process for epidemiologists, with Imperial College London and the WHO
- Tested which AI models found the most relevant research, setting the baseline the team tuned against
- Prototyped automatic collection of academic papers and their key details, replacing manual gathering

Bedarra Corporation | *Student Researcher* Aug 2024 - Oct 2024

- Built an AI assistant that answers questions about rarely documented programming languages
- Compared retrieval approaches to find which returned the most useful answers when documentation is scarce
- Wired the assistant to pull real code examples and documentation into every answer

Health Canada | *Data Scientist Intern* Sep 2023 - Dec 2023

- Built models that predict how toxic a chemical is likely to be, supporting risk assessments
- Built a pipeline that maps the relationships between chemicals, diseases, and published research
- Rebuilt the system that gathers chemical data, more than doubling usable sources with **100%** test coverage

Projects

Orbital Refuelling Simulator | *Python, Streamlit, scikit-learn, Qwen LoRA, pytest* github.com/nbui23/orbital-refuelling

- Simulated a **410-second** spacecraft refuelling mission across **9** phases, with **9** kinds of failure to catch
- Combined fixed safety limits with machine learning that flags subtler drift, keeping the two clearly separate
- Compared rule-based alerts against machine learning scores across every failure scenario
- Fine-tuned a small AI model to explain each alert in plain language, scoring **4.86/5** on held-out tests

Skills

Languages: Python, TypeScript, JavaScript, C++, Java, SQL
Frameworks & Runtimes: FastAPI, Flask, Streamlit, LangGraph, LangChain, Next.js, Node.js, Spring Boot
AI / ML / Data: LLMs, RAG, LangGraph, LangChain, Ollama, Qwen LoRA, scikit-learn, pandas, NumPy, PyTorch, FAISS
Databases & APIs: PostgreSQL/PostGIS, pgvector, MySQL, SQLite, MongoDB, Firebase, Neo4j, GraphQL
Tools & Practices: Git, Docker, Docker Compose, AWS, CI/CD, Ubuntu, TDD, pytest, unittest, Postman, Agile, SDLC
Spoken Languages: English, French, Vietnamese